Transit Score

Screening Model for Evaluating Community Suitability for Transit Investments

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The transit score is a measure that can inform the selection of appropriate transit investments for a given community. The model estimates a measure of the potential for success of various transit investments—the transit score—as a function of the densities of population, employment, and carless households. A version of this methodology was developed in 1989 by the Delaware Valley Regional Planning Commission (DVRPC), the metropolitan planning organization for the Philadelphia, Pennsylvania, and Camden–Trenton, New Jersey, region. The method was adapted and expanded by the New Jersey Transit Corporation (NJ TRANSIT) in 2000 to evaluate future rail and fixed guideway extensions in its 2020 plan. Several other versions of the transit score methodology are known to exist and have been applied in studies across the United States. In 2005, DVRPC staff, in collaboration with NJ TRANSIT, undertook a project to validate and apply the transit score methodology to areas within the DVRPC region. Three versions of the transit score model were calibrated with linear regression models by using observed transit journey-to-work mode share as a proxy for transit score. Development of the transit score model is documented, revisions and statistical validation are described, and a range of applications of the transit score, both completed and contemplated, are discussed.

A significant change is being seen in decision making for urban transportation investments in many large U.S. cities. Urban transportation planners are turning more toward investments in new transit lines and expanded services to relieve congestion and provide additional capacity to sustain new growth. Increasingly, local elected officials and developers press for the construction of new transit lines and the expansion of service on existing lines.

It has long been known that transit ridership is directly proportional to urban density (*1*). Yet not all developers and local officials are aware of this linkage or that transit is not as effective at attracting commuters out of autos when forced to operate in low-density environments. Consequently, in the face of pressure to expand transit service into numerous communities in New Jersey and in southeastern Pennsylvania, the Delaware Valley Regional Planning Commission (DVRPC) and the New Jersey Transit Corporation (NJ TRANSIT), New Jersey's statewide transit operator, sought ways to simply and graphically show how the consequences of land use planning and zoning decisions would impact the prospects for success of proposed new transit lines.

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Transportation Research Record: Journal of the Transportation Research Board, No. 2063, Transportation Research Board of the National Academies, Washington, D.C., 2008, pp. 115–124.
DOI: 10.3141/2063-14

BACKGROUND

DVRPC History of Transit Score

The present analysis has its roots in a July 1989 DVRPC report (*2*). That project identified locations as viable for transit service on the basis of three sequential factors: total work destinations, density of work destinations, and total work origins for employment centers above a certain threshold. Two later DVRPC studies refined this method of estimating transit viability (*3, 4*).

Variables were assigned threshold densities beyond which transit was presumed to be supported, and a given area was scored on each variable by calculating its local value as a proportion of the standard threshold. For example, the threshold for population density was 3.0 persons per acre (1.0 acre = 0.405 ha). If a given zone had a density of 6.0 persons per acre, its score for the population variable was 2.0. A zone's values (or scores) for each of the origin (residential trip-end) factors were averaged and then added to its score for employment density to arrive at its composite transit potential score. A composite score of 1.0 was assigned as being the minimum threshold for transit service viability. A summary equation is as follows (all densities gross):

$$\begin{aligned}\text{transit potential} = {} & [(\text{population per acre}/3.0) + (\text{households per acre})] \\ & + [(\text{0-car households per acre}/0.5) + \text{1-car households per acre}] \\ & + [(\text{seniors}[65+]\text{per acre}) + (\text{youths}[12\text{–}18]\text{per acre}/2.0)]/4 \\ & + [(\text{jobs per acre}/2.5)]\end{aligned}$$

Although each of the thresholds (for each input variable as well as for the composite score) was estimated logically, the actual relative significance of each input in affecting transit ridership was not assessed in any rigorous way.

NJ TRANSIT Use of Transit Score

In 2000, NJ TRANSIT had assembled a list of 34 transit projects under consideration throughout the state. The total estimated cost

for all the projects ran into billions of dollars. In addition, there were numerous proposals for new bus transit services, including shuttles to rail stations. Clearly, there would not be sufficient funding for all to proceed simultaneously. Priorities needed to be established for the projects on the list, and a way was needed to explain the logic for advancing some projects ahead of others to a variety of audiences, in an often politically charged atmosphere.

Consequently, NJ TRANSIT's business planning division developed a simple methodology to relate major new transit capital investments and new or expanded transit services to existing or forecast demographics throughout New Jersey. A statewide map displayed all the potential projects and transit scores for each census tract in the state for the year 2020. The relationships developed in DVRPC's described earlier work were modified and incorporated, as was earlier work by Pushkarev and Zupan on densities and volumes needed to support different types of transit service (*1*). Intermodal services, new bus services, and fixed guideway investments were included along with rail transit. The transit score was related to specific criteria that must be met, fully or conditionally, to qualify for further study.

The map, 2020 TRANSIT Possibilities for the Future, was published in a report to the public on future directions and challenges for the agency (*5*). The 2020 Transit Map, as it became known, clearly showed how projects aligned with expected urban growth and density patterns. The transit score methodology also was used in various projects as a sketch planning tool, especially for new fixed guideway and shuttle transit services.

CALIBRATING NEW TRANSIT SCORE MODEL

In 2005, DVRPC staff conceived a project, Creating a Regional Transit Score Protocol, in collaboration with NJ TRANSIT staff. This project was composed of two phases: evaluation of NJ TRANSIT's transit score methodology for use in Pennsylvania, followed by application of the selected method to evaluate current services and prospects for future services in the entire region (*6*).

Earlier work by both DVRPC and NJ TRANSIT had relied entirely on the judgment of the analysts in selecting the variables and assigning weights (coefficients) for the transit score model. Consequently, it was decided to try calibrating the transit score model by using empirical data. Calibrating the transit score equation could validate the underlying hypotheses and calculate the relative weights for the variables in the model.

A series of linear regressions was performed by using the transit score input factors as independent variables. Transit mode share for work trips, derived from the 2000 census, was used as a proxy for transit score and became the dependent variable. The geographic area for the model calibration was the combined universe of 2,934 tracts used for the 2000 census in the state of New Jersey and in DVRPC's five Pennsylvania counties (Bucks, Chester, Delaware, Montgomery, and Philadelphia).

Initial Transit Score Equation

The five input variables used in the previously existing transit score method were the initial set of independent variables. Each of the analysis densities was calculated on the basis of the gross land area of each census tract: population density (POPpACRE), housing unit density (HUNITpACRE), zero-car household density (0CARpACRE), one-car household density (1CARpACRE), and destination-based job density (JOBSpACRE).

In addition, a binary variable (HALFMI_RAIL) was assigned to each census tract on the basis of whether any portion of the tract was located within 0.5 mi of a rail station or fixed rail stop (subway, commuter rail, or light rail), to assess the role of proximity in transit viability. A similar variable relating to bus service proximity was also used. The density of bus service (BUSSTOPSpACRE) was calculated on the basis of the total number of bus stops for fixed-route service within the tract. If a given stop location was shared by five separate routes, that stop was counted five times in the density calculation. The values for each of these transit proximity variables (HALFMI_RAIL and BUSSTOPSpACRE) were calculated by using geographic information system (GIS) software and the most recent rail and bus layers provided by NJ TRANSIT and the Southeastern Pennsylvania Transportation Authority (SEPTA).

Values for the median travel time from (OrigMedianTravTime) and to (DestMedianTravTime) each tract were extracted from the Census Transportation Planning Package 2000. Finally, to supplement destination-based job density (JOBSpACRE), values for origin-based worker density in a tract (WORKERSpACRE) were extracted.

A weighted combination of origin- and destination-based transit mode share (AvgTRANSIT_SHARE×100) was calculated as the proportion of transit origins and destinations (combined) to total (all modes) origins and destinations. The specific values represent decimal transit mode shares multiplied by 100. When calculated independently, origin-based and destination-based transit mode share had a fairly high correlation (0.68) for the data set.

Correlation analysis found that POPpACRE, HUNITpACRE, and WORKERSpACRE were highly correlated (with correlation coefficients of 0.93 or 0.95). As the relationship between employed persons and total population or total population and housing units can be expected to be fairly consistent across census tracts, each of these factors effectively serves as a proxy for the other two. Consequently, only one of these independent variables, population density (POPpACRE), was selected for use in the regression analysis.

Table 1 summarizes the results of the initial regression analysis. As shown, many of the independent variables have a significant and positive relationship with the dependent variable. Job density and one-car household density were exceptions to this pattern; job density was determined not to have a significant relationship with transit mode share for the combined bistate data set, and one-car household density was calculated to have a significant but inverse relationship.

Revised Transit Score Equation

Because job density and density of one-car households were found to be not significant in explaining transit mode share, the model was recalibrated with both these variables removed. The revised transit score model is stated as follows:

$$\begin{aligned}\text{transit score (revised)} = {} & [0.13 * (\text{population per acre})] \\ & + [1.21 * (\text{zero car households per acre})] \\ & + \left[0.36 * \begin{pmatrix}\text{median travel time for JTW trips} \\ \text{originating in each area}\end{pmatrix}\right] \\ & + \left[0.34 * \begin{pmatrix}\text{median travel time for JTW trips} \\ \text{terminating in each area}\end{pmatrix}\right] \\ & + [2.47 \text{ if a rail station is located within 0.5 mi}] \\ & + [14.99 * (\text{bus stops per acre})] \\ & - 11.58[\text{constant}]\end{aligned}$$

where JTW is journey to work.

TABLE 1 Regression Summary, Initial Full Model

	Coefficient	Standard Error	*t*-Statistic	*P*-Value	Significant (Y/N) Positive/Negative
Intercept	−11.44	0.59	−19.39	.0000	
POPpACRE	0.19	0.02	9.30	.0000	Y, Positive
JOBSpACRE	0.01	0.01	1.37	.1703	N
0CARpACRE	1.18	0.08	14.55	.0000	Y, Positive
1CARpACRE	−0.25	0.07	−3.63	.0003	Y, Negative
OrigMedianTravTime	0.37	0.02	19.58	.0000	Y, Positive
DestMedianTravTime	0.34	0.02	15.79	.0000	Y, Positive
HALFMI_RAIL	2.39	0.23	10.34	.0000	Y, Positive
BUSSTOPSpACRE	14.07	1.15	12.25	.0000	Y, Positive
	df	SS	MS	*F*	Significance
Analysis of Variance					
Regression	8	217,941.5	27,242.7	793.2	0
Residual	2,925	100,459.3	34.3		
Total	2,933	318,400.9			
Summary					
Multiple *R*	0.827				
R^2	0.684				
Adjusted R^2	0.684				
Standard error	5.860				
Observations	2,934				

NOTE: Y, yes; N, no; df, degrees of freedom; SS, sum of squares; MS, mean square.

Recalibration for the revised model produced the following statistics: adjusted $R^2 = 0.682$ and *t*-stats POPpACRE = 9.55, 0CARpACRE = 15.30, OrigMedianTravTime = 19.47, DestMedianTravTime = 16.35, HALFMI_RAIL = 10.70, and BUSSTOPSpACRE = 14.24.

Calibrating Simplified Final Transit Score Model

Although the validity of the full transit score equation was enhanced by the inclusion of factors such as median travel time and bus stop density, such factors are difficult to estimate without extensive data resources. For these reasons, it was determined that a second, simpler transit score equation should be validated by using the same regression method but with a subset of the original independent variables. The resulting equation includes only those independent variables that can be readily estimated for a given plan or scenario and does not depend on the presence of existing transit service. Specifically, the variables tested in the regression analysis for the simplified transit score method were as follows:

dependent variable: AvgTRANSIT_SHARE × 100

independent variable: POPpACRE

JOBSpACRE

0CARpACRE

The density of zero car households is included, but the density of one-car households is not. This is because the number of zero-car households reflects a population almost entirely dependent on transit whereas one-car households have other alternatives. It was determined that the simplified transit score equation should have no constant (i.e., the regression was conducted with the constant suppressed). The resulting model appears more intuitive—no unexplained extra factor will be seen to be affecting scores, and if population and job densities are zero, transit mode share also should be zero. Table 2 summarizes the results of the regression.

Each of the independent variables was found to have a significant and positive relationship with transit mode share. The significant and positive result for job density is notable, as this variable was not found to be significant in the full regression for the same data set. On the basis of the coefficients calculated during the regression analysis, the simplified transit score equation (with the transit score simply substituting for transit mode share) is as follows (all densities reflect gross land acres)

$$\begin{aligned}\text{simplified finaltransit score} = &[0.4 * (\text{population per acre})] \\ &+ [0.09 * (\text{jobs per acre})] \\ &+ [0.74 * (\text{zero car households per acre})]\end{aligned}$$

SETTING INTERVALS FOR FINAL TRANSIT SCORE MODEL

Once the final coefficients had been estimated for the transit score equation, the next step was to classify the range of scores into homogeneous intervals corresponding with levels of transit investment appropriate for the various geographic subdivisions of the region. An initial approach was to divide the range of estimated transit scores into equal quintiles and display the results on a GIS map that included all of New Jersey plus the DVRPC region's five Pennsylvania counties. The resulting map showed scores for some outlying areas

TABLE 2 Regression Summary, Simplified Final Model

	Coefficient	Standard Error	*t*-Statistic	*P*-Value	Significant (Y/N) Positive/Negative
POPpACRE	0.41	0.01	27.44	.0000	Y, Positive
JOBSpACRE	0.09	0.01	10.52	.0000	Y, Positive
0CARpACRE	0.74	0.10	7.74	.0000	Y, Positive
	df	SS	MS	*F*	Significance
Analysis of Variance					
Regression	3	397,150.3	132,383.4	2,278.922	0
Residual	2,931	170,262.9	58.09038		
Total	2,934	567,413.2			
Summary					
Multiple *R*	0.837				
R^2	0.700				
Adjusted R^2	0.700				
Standard error	7.622				
Observations	2,934				

that appeared to support higher levels of transit service than had been sustainable in practice. In addition, NJ TRANSIT staff began to compare the transit score analysis with state planning areas as designated by the New Jersey Office of Smart Growth. These are large areas of land that share a common set of conditions such as population density, infrastructure systems, level of development, or natural systems, as identified in the New Jersey State Development and Redevelopment Plan (*7*). Consequently, the initial five quintiles were adjusted to better correspond with both existing transit service and projected areas where growth would be consistent with state policy. The resulting categories are shown in Table 3 along with the distribution of several key demographic variables (*8*).

Transit score categories rated as medium and above correspond roughly with the areas the state plan targets as locations where growth should occur and where most transit service is viable. Per 2000 data, these areas

- Constituted 78.2% of the population of the state,
- Constituted 75.6% of the locations where workers reported to their primary work or employment,
- Had 77.1% of the households of the state, and
- Had 20.9% of the land area of the state.

As Table 3 shows, the two highest categories account for just over 50% of the state population and households on about 8.4% of the land area. These areas had about 46% of the employment. Employment is a bit more spread out into lower transit score categories, reflecting the more dispersed nature of New Jersey employment. Figure 1 is a map of the transit scores calculated by using 2000 census data for the state of New Jersey and the five Pennsylvania counties included in the DVRPC region.

INVESTMENTS COMMENSURATE WITH TRANSIT SCORE INTERVALS

The transit score indicates the relative likelihood and potential for different types of transit usage in a geographic area. Areas having higher scores can accommodate a greater range of types (modes) of transit service. The transit score is used to identify where various types of transit investments may be appropriate, subject to available resources, provided certain criteria and conditions are met (*9*). Three overall categories of transit investments were defined. Together with some additional decision criteria, they are as follows:

TABLE 3 Transit Score Category Ranges and Associated Variables for New Jersey (Year 2000)

Transit Score Category	Transit Score Range	Population	Primary Employment	Households	Land Area
High	>7.5	23.4%	16.9%	22.8%	1.5%
Medium-high	2.5 to 7.5	31.0%	29.4%	31.6%	6.9%
Medium	1.0 to 2.5	23.8%	29.3%	23.7%	12.5%
Marginal	0.6 to 1.0	6.5%	9.5%	6.8%	7.0%
Low	<0.6	15.3%	14.9%	15.1%	72.1%
Total		8,414,000	3,962,000	3,310,000	7,418 mi^2

NOTE: 1 mi^2 = 2.59 km^2.

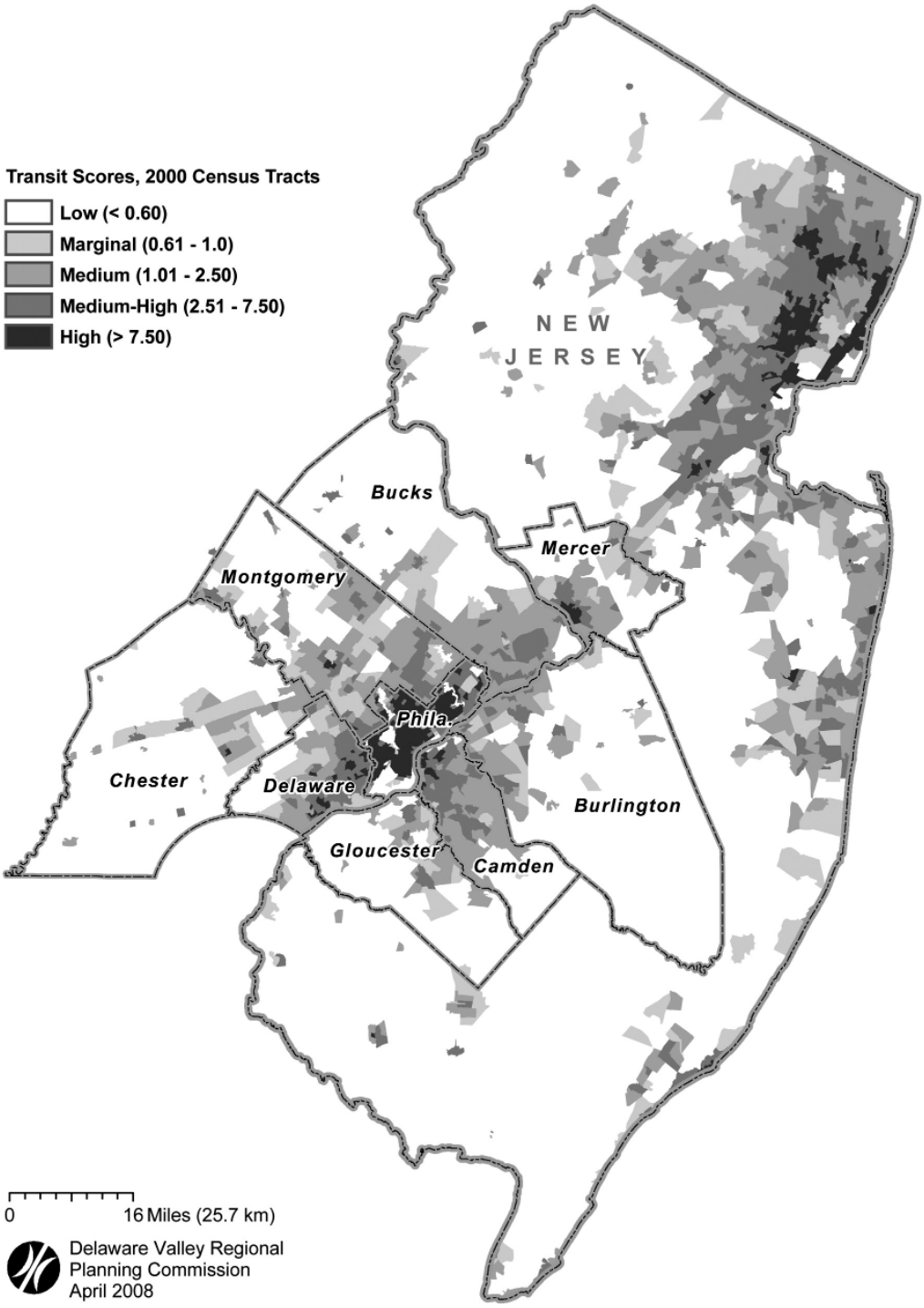


FIGURE 1 Transit scores for New Jersey and DVRPC five-county Pennsylvania region.

- Fixed guideway transit (commuter rail, light rail, ferry, bus-only lane)—new transit lines and extensions of existing lines. Fixed guideway transit requires significant capital investment and is primarily on its own right-of-way with no or limited mixing with auto traffic. Fixed guideways often can provide time savings compared to auto travel. Each type of fixed guideway project must meet certain minimum criteria, primarily related to having at least part of the line or service in an area with a high transit score and meeting a minimum threshold for the number of jobs in the center or municipality. Other criteria include availability of right-of-way and adjacent area transit scores.
- Bus service potential (includes express bus, local bus, minibus service). Levels of bus service are related to the transit score of an area, with a range of daily spans of service and a range of average hourly frequencies of bus service. For some services, a minimum number of jobs in a municipality or center is required, along with variations based on location of an area in the state plan.

• Intermodal or access to transit (includes park–ride, shuttles to transit, intermodal terminals). These are services and projects that provide access to transit service and facilitate intermodal or multimodal service. On the basis of the transit score, peak period ridership, and other factors, minimum guidelines are outlined for park–rides, shuttle buses, and other intermodal facilities such as parking structures and terminals.

Table 4 summarizes for each of the three transit investment categories the modes, services, and intermodal facilities that meet demographic and transportation criteria and are applicable for implementation as based on a geographic area's transit score. Y (yes) means the service or fixed guideway mode meets all conditions and is an appropriate investment if an area is classified as having the transit score category indicated. C (conditionally appropriate) means that the appropriateness of the service or investment is less certain, but the project may be appropriate depending on demonstrated trip patterns and other factors, such as number of jobs at a destination. A dash means that service or fixed guideway mode is not warranted or is not an appropriate investment for that transit score category.

It needs to be stressed that if the transit score and other criteria or conditions are met and a specific fixed guideway mode or transit service appears to be an appropriate investment, a project or service can advance to more detailed feasibility studies. The transit score indicates the order-of-magnitude relationship for types of fixed guideway, bus service, and intermodal investments. However, more detailed studies are necessary to document ridership estimates, costs, benefits, environmental and engineering feasibility, and financial impact. These studies will determine if the investment types outlined in the transit score are feasible. Conversely, not meeting the transit score criteria does not automatically indicate a project has no merit, but it does mean that the project fails to meet even a minimum level of justification. More

TABLE 4 Summary of Transit Score Applicability for Transit Investments by Area

	Transit Score Category				
Transit Investment Type	High	Medium-High	Medium	Marginal	Low
Fixed Guideway by Type					
Rapid transit	C	—	—	—	—
High capital cost electric light rail (LRT)	C	—	—	—	—
Medium-low capital cost electric LRT	C	C	C	—	—
Commuter rail terminal	C	—	—	—	—
Commuter rail or diesel LRT	Y	Y	C	C	—
Monorail or personal rapid transit (PRT)			*Special case*		
Ferry	Y	C	C	—	—
Recreational transit	C	C	C	—	—
Bus lanes–limited access roads	Y	C	—	—	—
Bus lanes–arterials	C	—	—	—	—
Bus priority treatment	Y	C	C	C	—
Bus Service Potential and Other Transit					
Express bus as a destination or terminus	C	—	—	—	—
Express bus–walk only access	Y	Y	—	—	—
Express bus–park and ride access	—	Y	Y	Y	Y
High-intensity local bus service	Y	—	—	—	—
Medium-intensity local bus service	Y	Y	—	—	—
Minimum-intensity local bus service	—	Y	Y	C	—
Local circulator bus–rural center	—	Y	Y	C	C
Local social service or paratransit	Y	Y	Y	Y	Y
Minibus with line-haul transit	C	Y	Y	C	—
Minibus express suburban service	Y	C	C	—	—
Vanpools and vanpool subsidy	C	Y	Y	Y	Y
Intermodal and Access to Transit					
Shuttle bus to line-haul transit (walk access)	Y	C	C	—	—
Shuttle bus to line-haul transit (remote parking)	—	—	C	—	—
Structured park and ride	C	C	—	—	—
Surface park and ride for rail–LRT–ferry	C	Y	Y	Y	—
Multimodal terminals	Y	C	—	—	—

NOTE: Y = Yes, applicable with no conditions. C = Conditional, applicable if conditions are met. — = Not applicable or warranted.

TABLE 5 Example Criteria for Conditionally Appropriate Transit Investments for High-Score Areas

Rapid transit	Only if direct connection to Philadelphia or Manhattan or 15,000+ jobs in center.
Commuter rail	As a destination or terminal—only if serving a regional center with 60,000+ jobs in municipality.
High capital-cost electric light rail	33% of line in tunnel or elevated. Must serve 30,000+ jobs in center or municipality, 60,000 jobs preferred.
Medium and low capital-cost electric light rail	Must serve 30,000+ jobs in center or municipality to be line terminal.
Bus priority treatment	On major arterials with 40+ buses/peak hour direction.
Bus-only ramps and lanes	On limited access roads and connectors to regional centers with 60,000+ jobs.

detailed investigation or special circumstances can determine that a service or guideway project is viable outside the transit score criteria.

Table 5 elaborates on the entries shown in Table 4 for "conditional, applicable if conditions are met" considerations for fixed guideway investments that may be appropriate in areas attaining a high transit score. These conditions were developed specifically for New Jersey and could be modified for other areas.

APPLICATIONS OF TRANSIT SCORE MODEL

Transit Score Comparison for DVRPC Region, 2000 to 2030

The transit score will have broad applications from a regional planning standpoint, including a refined use in DVRPC's congestion management process and applications in forthcoming long-range plans. The most straightforward application is a simple comparison between present-day transit scores across the region (reflecting 2000 census data) and transit scores reflecting DVRPC's 2030 regional growth projections. Figure 2 presents this comparison, depicting transit scores at the level of traffic analysis zone.

One conclusion is that the transit-supportive areas in the region are generally concentrated among Philadelphia and its inner-ring suburbs and that existing transit infrastructure is generally collocated with these supportive areas. Generally speaking, areas that are forecast to add population and employment of magnitudes that would push them into a higher transit score classification occur adjacent to other areas that are already transit supportive.

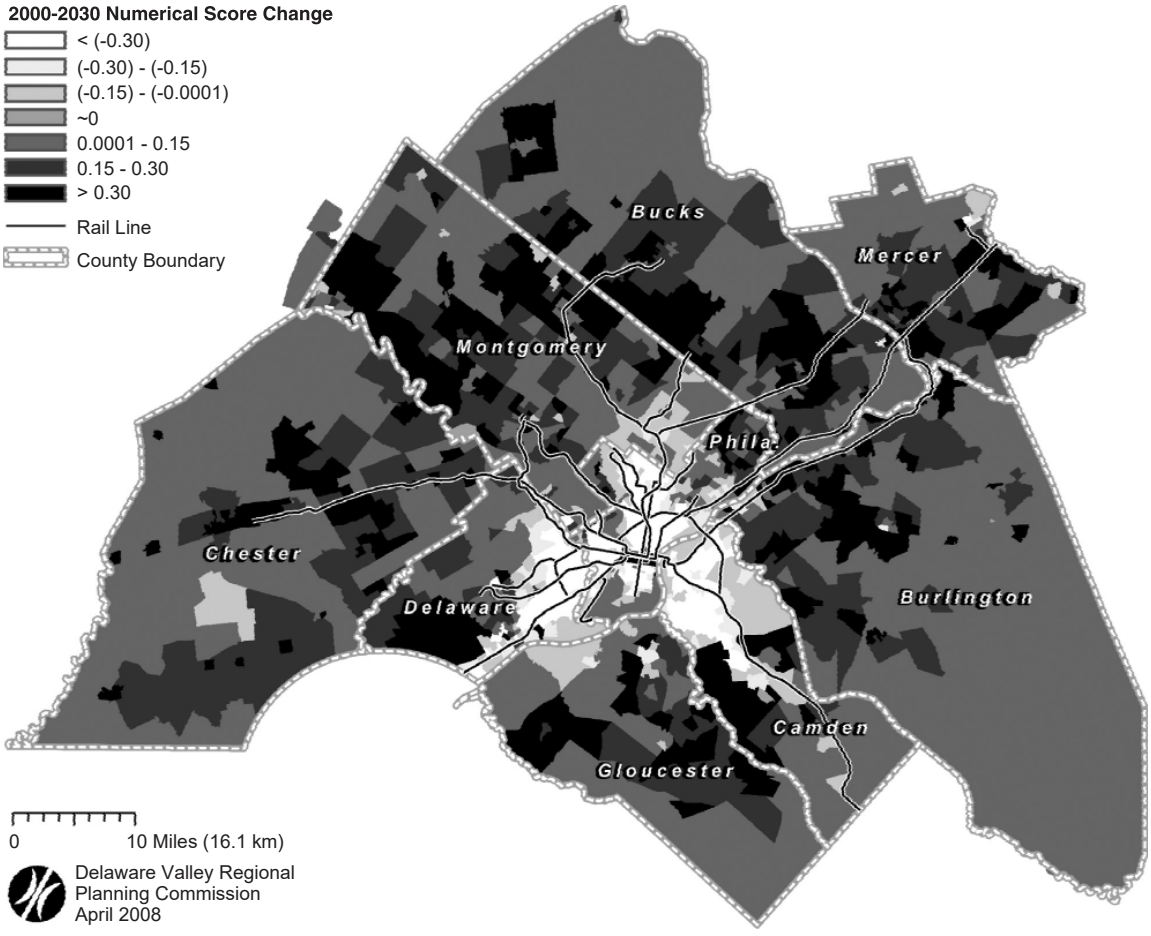


FIGURE 2 Projected numerical transit score change, 2000 to 2030, DVRPC region.

Build-Out Comparison for US-322 Corridor in New Jersey

DVRPC recently completed a multiyear planning project for the US-322 corridor in Gloucester County, New Jersey. Phase 1 of this effort included build-out analyses for each of four study area municipalities: Logan Township, Harrison Township, Swedesboro Borough, and Woolwich Township. These analyses resulted in calculations of the population and job growth that would result from full build-out under prevailing zoning, including any zoning adjustments resulting from active redevelopment plans (*10*).

As part of the project's second phase, the transit score was used to illustrate the variation in transit supportiveness that would result from an even distribution of this build-out growth across each municipality in comparison with concentrating all build-out growth in a series of conceptual centers, each 2 mi in diameter (*11*). The specific method for each exercise is summarized in Figure 3, which compares sprawl and smart growth scenarios. The comparison shows that by concentrating the growth in transit supportive centers rather than in an even distribution, additional transportation options become viable. In this example, an east–west bus rapid transit line along the corridor could be supported. This expansion of mobility options makes a per capita reduction in auto trips, and traffic, possible.

New Jersey Office of Smart Growth

The New Jersey Office of Smart Growth, housed in the New Jersey Department of Community Affairs, ensures that municipal, county, regional and state agency plans are consistent with the State Development and Redevelopment Plan and with each other. This is accomplished through a voluntary process called plan endorsement. An endorsed plan entitles municipalities and counties to a higher priority for available funding, streamlined permit reviews, and coordinated state agency services (*12*).

The Office of Smart Growth has prepared draft regulations for revising the plan endorsement process that would include preparation of a transit score analysis as part of a state agency assessment report and action plan. The state agency report reviews a petitioner's plan and proposed planning and implementation agreement for consistency with the goals, policies, and strategies of the state plan. Use of the transit score analysis would identify the role of transit in the plan endorsement process and enable municipalities to identify the potential of transit for future land use alternatives (*13*). The Office of Smart Growth will prepare a document describing the transit score, and NJ TRANSIT has agreed to provide training to the Office of Smart Growth in performing transit score analysis.

Other Uses of Transit Score

In addition to the preceding examples, there is evidence that the transit score has been used in several other studies. Urbitran Associates used a version of the transit score, called the transit success score, in preparing a 2006 transit development plan for the city of Lincoln, Nebraska (*14*). Mercer County, New Jersey, in its master plan update conducted in the fall of 2006, used an earlier version of the transit score to map locations in the county that have the greatest potential for new transit infrastructure investments (*15*). The New Jersey Meadowlands Commission had proposed to use the transit score in its draft 2007 Meadowlands District transportation plan (*16*).

LIMITATIONS AND FUTURE RESEARCH

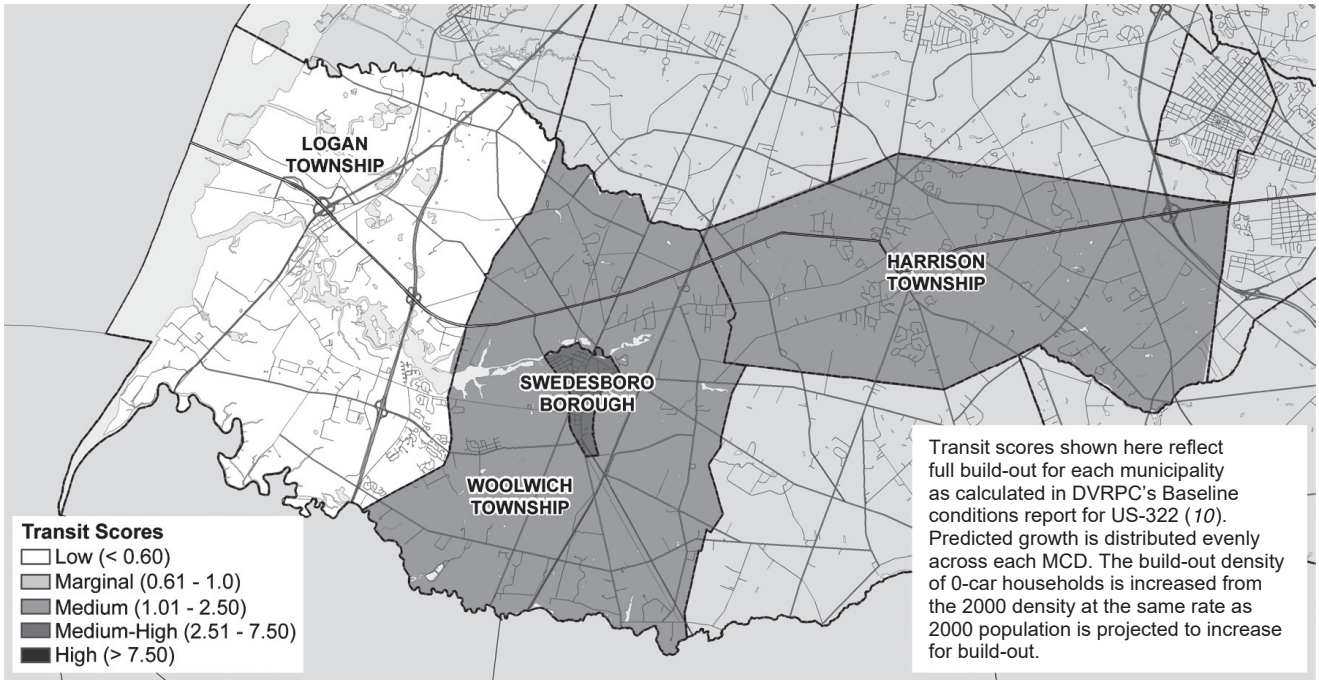
Completion of the DVRPC transit score project marks the end of an initial phase of research into utility of the transit score. It is proposed as a method for evaluating whether the existing or proposed land use densities in a given community are likely to support certain types of transit investments, as based on modes splits extracted from the most recently available census journey-to-work data. As such, the transit score should not be used as the only indicator to predict the likely success of a transit investment. There are other factors that could lead to a successful transit project in an area with a low transit score. Among these factors are the degree to which travel markets reflect concentrations of trips at both the origin and destination ends and the network conditions affecting travel in a given market, such as congestion and speed.

In this context, there are a number of opportunities to explore further the usefulness of the methodology and to improve its accuracy and predictive ability. One aspect of the score that would benefit from additional research is the use of zero-car household density as one of the variables. Whereas density of population and employment are subject to regulation by zoning, and can be influenced by public policy decisions, the locations of households without autos is not likely to be policy driven in contemporary American society. However, there may be other policy-driven variables that could be examined for use in a predictive context, such as the presence of low-income housing units or housing for elderly residents.

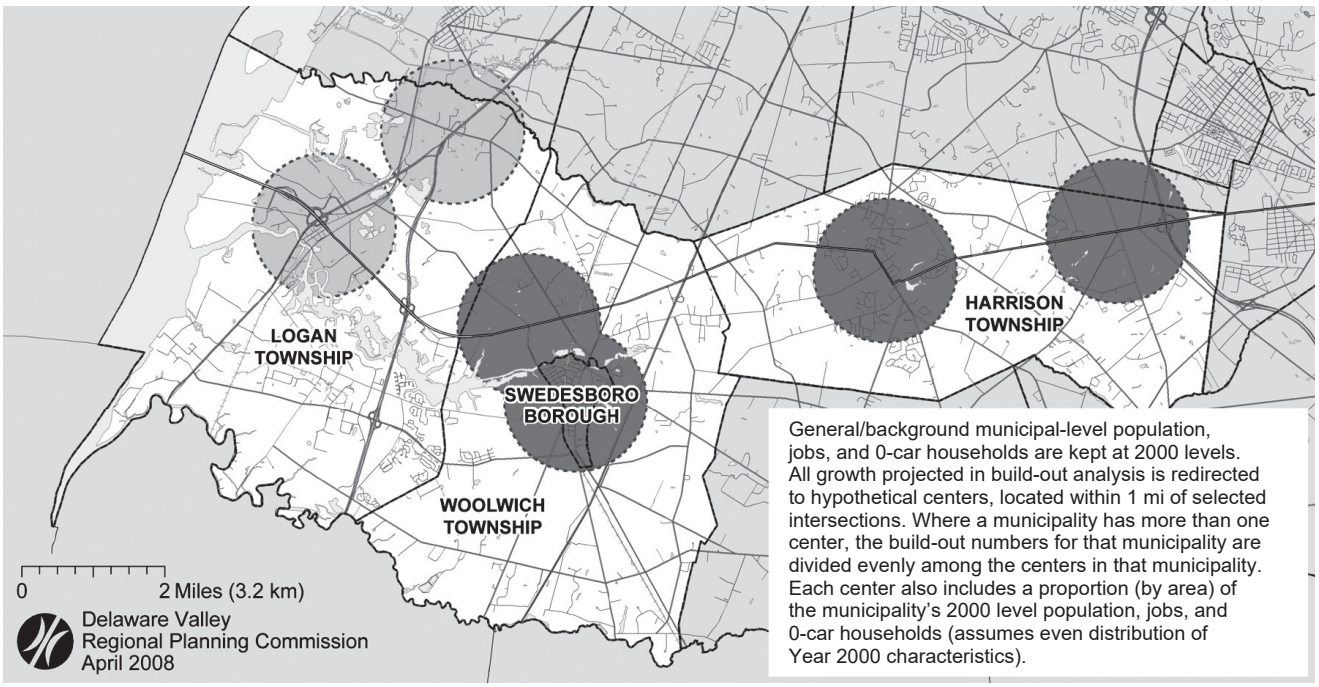
A second issue that may warrant additional research is the expansion of the transit score to a corridor-level analysis scale. NJ TRANSIT links transit scores along corridors in considering new transit investments and services. However, this has not been tested extensively, and further research is warranted.

As noted in the sections on model calibration, the journey-to-work mode split data from the most recent (2000) census was used as a proxy for transit score. Work trips, which account for 80% of NJ TRANSIT riders, are becoming a smaller fraction of total trip making. As nonwork trips increase as a proportion of all trips, it would be useful to extend the transit score model beyond journey to work. It would be desirable to examine mode split for nonwork trips. In addition, mode split reflects the relative quality of the existing highway and transit systems and the distribution pattern for trips to other locations. Comparative attributes of the various transportation modes may not exhibit the same relationships in the future. Consequently, further research could identify better proxies for transit score.

Finally, the appropriateness of application and accuracy of the transit score depends to a large measure on the geographical scale at which it is applied. For the analysis used in the DVRPC and NJ TRANSIT applications, the geographic unit of analysis is the census tract, a unit of analysis that varies in area. Census tracts vary in size across a region from a few city blocks to an entire municipality. It is therefore possible for population and employment densities to be very heterogeneous across a large census tract. Consequently, the calculated transit score may reflect an average that masks very large differences within a geographic area, which could lead to inaccuracies in the transit score analysis. Further research may help determine ways to subdivide areas to create more homogeneous geographic subdivisions that increase the accuracy of the methodology.



(a)



(b)

FIGURE 3 Transit scores for US-322 corridor: comparison of scores at build-out (*a*) under prevailing zoning (MCD = minor civil division (sprawl scenario) and (*b*) under redirection of growth to proposed centers (smart growth scenario).

CONCLUSION

The work performed under the DVRPC transit score project provided a transit score model that, unlike the earlier versions described, is validated by empirical data for the state of New Jersey and the five-county Pennsylvania portion of the DVRPC region. This paper described how the model was calibrated by using work-trip mode split as a proxy for transit score. Three regression equations were documented, showing how the model evolved to a simplified three-variable version that can be readily applied in a variety of applications. A number of applications for the transit score model were documented and directions for further research were discussed.

On the basis of this research, the transit score model is proposed as a viable tool for performing screening assessments that can be used to select appropriate transit investments for a given community.

ACKNOWLEDGMENTS

This work was partially funded by FTA, passed through and supported by the Pennsylvania and New Jersey departments of transportation, with local financial matching provided by counties and cities in the southeastern Pennsylvania section of the DVRPC region. The authors acknowledge the contributions of the following individuals, who provided advice and support for the effort: Kurt Paulsen of the Department of Urban and Regional Planning at the University of Wisconsin–Madison, who consulted on statistics; and Richard T. Roberts of NJ TRANSIT, who reviewed the paper and provided helpful comments.

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The authors are responsible for the findings and conclusions, which may not represent the official views or policies of the funding agencies.

The Public Transportation Planning and Development Committee sponsored publication of this paper.